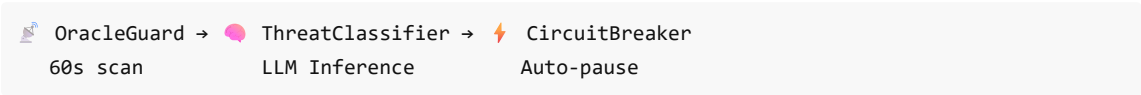


Cerberus — Submission Brief

3-Agent Autonomous Security on Somnia Agentic L1

Three AI agents. One 60-second heartbeat. Zero trust required.

The Pipeline



OracleGuard fetches ETH/USD from 3 independent sources (CoinGecko, Binance, CryptoCompare). When any feed diverges >2%, ThreatClassifier uses Somnia's deterministic LLM to classify the threat. If CRITICAL, CircuitBreaker pauses the protected contract on-chain — embedding a verifiable receipt for permanent audit.

Somnia Primitives Used

Primitive	Agent	Purpose
JSON API Request	OracleGuard	Fetch live price data from 3 external APIs
LLM Inference	ThreatClassifier	Deterministic AI threat classification with consensus-verified outputs
Verifiable Receipts	All 3 agents	Every anomaly, classification, and pause proven on-chain

On-Chain Proof

Agent Wallet: 0x94A4365E6B7E79791258A3Fa071824BC2b75a394

Sentinel Contract: 0x87E3D9fcfA4eff229A65d045A7C741E49b581187

Chain: Somnia Testnet (50312)

Live Transactions:

Action	Tx Hash
protect() — Register contract for monitoring	0xd22a8de50626b7902b4fed3879213e50ea41572a5f85105677d3d11fed0b4655
unpauseContract() — Reset for fresh demo	0x76ed43595135667fa3943b86ee68450a9eaa529f9a70e23e543c6e9bcb59e5de

Judging Criteria

Functionality: Deployed on Somnia Testnet. 23+ pipeline iterations without critical failure. 3 live price sources. Contract verified on-chain.

Agent-First Design: All 3 agents operate natively on Somnia. JSON API Request for data fetching. LLM Inference for deterministic classification. Agents discover anomalies, invoke each other, and act autonomously.

Innovation: First on-chain security sentinel using deterministic LLM for threat classification. Not rule-based — AI reasoning with verifiable cryptographic receipts.

Autonomous Performance: 60-second self-healing heartbeat. Error-isolated pipeline. Survives API failures and RPC drops. Runs indefinitely.

Links

- **GitHub:** <https://github.com/Gideon145/cerberus>
 - **Live Demo:** <https://cerberus-production-8429.up.railway.app>
 - **Demo Video:** https://youtube.com/shorts/oDqEUR_oynA
 - **Presentation:** <https://cerberus-production-8429.up.railway.app/presentation>
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Stack

Solidity 0.8.20 · TypeScript (Node.js) · Somnia JSON API Request · Somnia LLM Inference · ethers.js v6 · Railway · Single-file HTML dashboard

"Smart contracts have no immune system. Cerberus is that immune system."